CANCER TREATMENT FRAMEWORK

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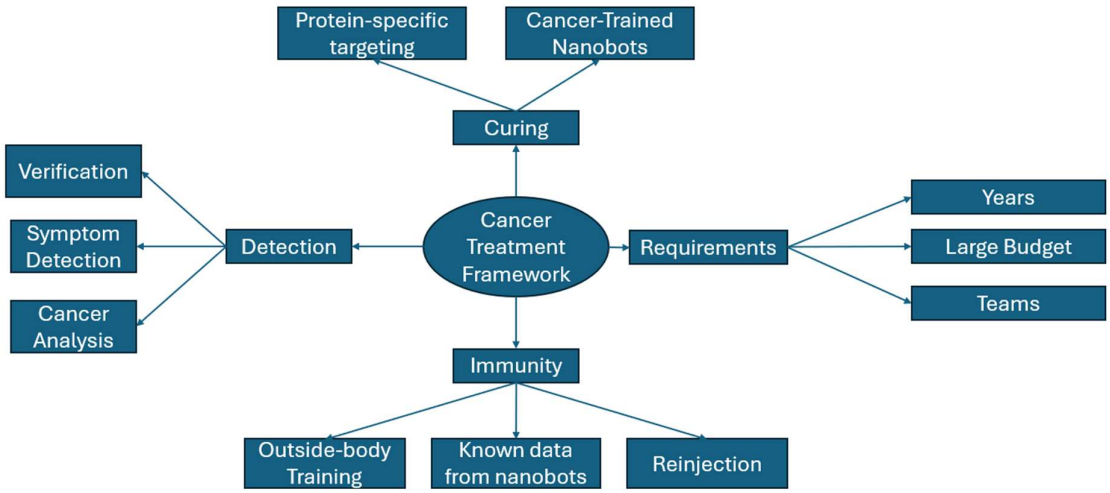
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Introduction

This cancer treatment framework was designed to aid medical professionals and doctors in curing cancer around the world. This was also designed to show the compatibility between technology and medicine. This cancer treatment framework was developed as cancer is one of the most dangerous medical diseases in humanity's history. This is because millions of people die every year due to cancer. This cancer treatment framework uses a mixture of human biology and nanotechnology to treat and even build immunity to cancer. This is just a design, and it will mostly likely take years to fully develop this framework due to the medical and technological advances needed. However, I believe this high cost and time-consuming process is more than worth it to save millions of people's lives.

This cancer treatment framework cannot promise perfect success rates against cancer or full immunity to it. However, this cancer treatment framework can promise to reduce the effectiveness of cancer and provide immunity within human limitations. It can also promise to provide a consistent approach to detecting, curing and building immunity to cancer. It is important to remember that to fully develop to this cancer treatment framework it will require a team of nanotechnologists and cancer researchers to do so. Make sure to adapt this framework to the specific type of cancer and real medical results.



The spider diagram above shows a visual layout of the cancer treatment framework. This shows the key features of the framework for detection, immunity and curing. It also shows the estimated requirements of full developing this framework. More details on the process can be found below.

Initial Cancer Detection Layer

- **It is important that medical professionals can detect the signs of cancer and analyse it before it reaches a terminal stage.**
- **This framework offers a 5-step process for detecting cancer and increasing cancer awareness.**
- **The first step is cancer communication.**
- **This step is where hospitals and medical professionals clearly warn and document the symptoms of cancer online for people to understand.**
- **This step can be done by using communication channels like social media or medical forums online to increase cancer awareness.**
- **Medical professionals can also use the structure What (What is the specific cancer?) → Why (Why does it happen?) → How (How does it happen?) → Common symptoms (What are the common symptoms?) → Advice (What can you do if you are experiencing these symptoms?).**
- **This may be in paragraph or bullet point format for easier and quicker understanding by patients or the population in general.**
- **The second step is symptom detection**
- **This step is where medical professionals check the symptoms against known cancer symptoms or other known diseases.**
- **This step can be done by checking for prior medical infections or issues and checking for any internal or physical symptoms that the patient may have.**
- **This step is important to understand if the patient is dealing with cancer or another medical issue to ensure that they can be applied treatment based on the issue.**
- **The third step is verification.**
- **This step is where medical professionals verify that the patient has cancer before beginning cancer analysis or cancer-specific treatments.**
- **This step can be done through screening, blood tests, urine testing, x-ray, and other cancer-specific detection methods.**
- **The fourth step is cancer analysis**
- **This step is where medical professionals analyse what stage the cancer is in, the type of cancer and its current infection speed.**
- **This step can be done through testing for tumour markers based on an estimate with the symptoms, further screening for where the mass is located and tissue sampling where the tissue is cross-referenced against known biological cancer tissue structures.**
- **The fifth step is documentation**
- **This step is where medical professional document the results of their cancer analysis into a structured and consistent format.**
- **This step can be done by documenting the type of cancer, the location of the mass, the biological structure of the cancer cell and any medical issues.**
- **This step is important because this information will be used for the nanobots.**

Cancer Treatment Curing Layer

- **It is important that medical professionals not only detect cancer but also be able to provide treatment for it.**
- **This framework offers a 6-step process in curing cancer with nanotechnology.**
- **The first step is nanobot training**
- **This is where nanotechnologists and medical professionals work together to train the nanobots.**
- **This can be done by using the machine-learning, training data based on the documentation of cancer on the patient and running practical tests against cancer cells outside the body**
- **The second step is nanobot verification**
- **This is where nanotechnologists and medical professionals check the nanobots for technical issues or if the patient's immune system may reject the nanobots.**
- **This can be done by checking prior medical history for allergies and running diagnostics tests on the nanobots.**
- **The third step is nanobot deployment**
- **This is where medical professionals deploy the nanobots into the patient body via targeted injection.**
- **This can be done by using the needle to make a hole in the skin for the nanobots to enter the body through successfully.**
- **The fourth step is cancer destruction**
- **This is where the nanobots find and destroy the cancer cells within the mass located site and other areas of the body if applicable.**
- **This is done by the nanobots using circulation for movement and protein-specific targeting for destroying cancer cells like antibodies**
- **This step is important as it makes sure that the immediate threat to the patient is destroyed and handled effectively.**
- **The fifth step is nanobot transmission**
- **This is where the nanobots transmit the cancer data to the remote system to be used for training outside of the body.**
- **This is done as the nanobots use a wave signal to transmit the cancer-specific data such as markers, DNA/RNA info and cellular structure to the system.**
- **This step is important to allow medical professionals to recreate cancer cells and train the patient white cells for immunity against that specific cancer.**
- **The sixth and final step is nanobot disposal**
- **This is where the nanobots disposal of themselves after transmission to prevent any chances of immune response to them or medical issues.**
- **This is done as the nanobots dissolve inside of the body due to their polymer material and exit via the body's digestive processes.**
- **This step is important as the nanobots are not designed to stay in the body for a long period of time and may cause blockage if they were to get stuck.**

Cancer Treatment Training Layer

- **It is important that not only does this cancer treatment framework provide detection and curing but also immunity.**
- **This framework offers a 5-step process for training white blood cells and giving the patient immunity against a specific cancer**
- **The first step is cellular recreation.**
- **This is where medical professionals recreate the cancer cells based on the data transmitted and collected from the nanobots.**
- **This can be done through protein synthesis, cell line engineering, validation and expansion which all can be used to recreate the cancer cells.**
- **This step is important as it allows T-cells to have something to train against.**
- **The second step is white cells extraction.**
- **This is where medical professionals extract white T-cells from the patient body for training them.**
- **This can be done by leukapheresis, T-cell isolation and Activatio to safely and carefully extract the white blood cell.**
- **This step is important as the body is extremely unlikely to reject is own defence system.**
- **The third step is white cell training.**
- **This is where medical professionals train the T-cells to specifically destroy that specific-type of cancer cell.**
- **This can be done through co-culture, expansion and selection, testing for specificity and cryopreservation or immediate use.**
- **It is important that these T-cells are verified to ensure that they don't attack healthy cells within the patient body via testing for specificity.**
- **This step is important to teach the patient's body on how to handle and protect itself against that specific type of cancer building immunity.**
- **The fourth step is reinjection.**
- **This is where medical professional reinjects the T-cells back into the patient's body.**
- **This can be done through a targeted injection of the trained T-cells back into the patient's body specifically where the mass of the cancer cells was located.**
- **This step is important to ensure safe transition into the patient's body and to test the T-cells against any cancer cells that the nanobots may have missed.**
- **The fifth step is monitoring**
- **This is where medical professionals monitor the patient for any side-effect symptoms or any immune system rejection.**
- **This can be done through weekly or daily check-ups, screening, blood tests and urine samples to ensure that the patient isn't being affected for cancer.**
- **This step is important as some patients may experience intense side-effects or immune system rejection which may require intensive or separate treatment.**

Cancer Treatment Nanobot Design

- **This nanobot design is designed to be safety-first and reliable**
- **This framework focuses on 3 areas for this nanobot design.**
- **The first area is size and material.**
- **For material, the nanobots will be made from biodegradable polymer which can dissolve easily within the human body.**
- **This is important because the nanobots need to be able to exit the body safely without causing any harm to the patient.**
- **For size, the nanobots will be 3000 nanometres to be able to fit through Arteries, Capillaries, Arterioles, Venules and Veins.**
- **3000 nanometres was picked as the best size as it is still smaller than red blood cells and is large enough to fit the components needed for the bot.**
- **The second area is physical computing components**
- **For the processing, the nanobot will use an ASIC also used as an application-specific integrated circuit.**
- **This is important as the nanobot is too small for a general-purpose CPU which would require the nanobot to be larger than the red blood cells.**
- **For storage, the nanobot will use RRAM or MTJ for the temporary storage of data that is needed.**
- **This is important as the nanobots need to store the data they collect from the cancer cells somewhere until it can be transmitted.**
- **For sensors, nanobots will use biodegradable electrochemical sensors to detect the cancer cells within the body.**
- **This is important as nanobots must be able to find cancer cells specifically especially not to accidentally target healthy cells.**
- **For power, the nanobot will use a solid-state thin-film battery to power all the components and the systems used.**
- **This is important because the nanobot needs to last a few hours to perform its job correctly within the body.**
- **The third area is nanobot functions & systems.**
- **For data collection, the nanobots use their sensors to collect data on cancer markers and some surface-level data on the cancer cell itself.**
- **This is important as doctors need to train T-cells on what to target to avoid attacking health cells and for more accurate protection.**
- **For targeting, the nanobots use their sensors to detect the cancer cells, use ligands to bind onto the cancer cells, and then release 1V to kill the cancer cells.**
- **This is important as the nanobots need reliable, accurate and safe targeting.**
- **For self-destruction, the nanobots perform the action if the battery reaches 3% or if it is remotely destroyed by a signal.**
- **This is important as the nanobots must be able to self-destruct without remaining within the body for extremely long-durations.**

Cancer Treatment Patient Workflow

- **This workflow will show the full process of a patient detecting the symptoms of cancer, having it cured and then having full, long-term immunity.**
- **First, the patient is scrolling on social media and sees a post about lung-cancer awareness.**
- **They look at the common symptoms which include persistent coughing, chest pain, shortness of breath and coughing up blood.**
- **Second, the patient remembers they have been coughing persistently for a few days and have recently started having chest pain.**
- **The patient decides to book an appointment to their local clinic or GP for a check-up just in case.**
- **Third, the doctor cross-references her symptoms against known diseases in the database.**
- **The doctor from the results determines it is mostly likely to be either acute bronchitis or lung cancer.**
- **Fourth, the doctor books an appointment with the patient to see a pulmonologist to verify if the patient has lung-cancer.**
- **Fifth, the patient arrives to the pulmonologist who is prepared to do some tests to confirm if the patient can be diagnosed with cancer.**
- **The pulmonologist performs chest x-ray scans, biopsies and uses CT-scan to confirm the presence of lung cancer.**
- **Sixth, the pulmonologist confirms to the patient that they have been diagnosed with lung cancer and ask them for further appoints for cancer analysis.**
- **The pulmonologist performs further chest x-ray scans, biopsies and may use tissue sampling with comparing to known cancer infected tissues to analyse the cancer.**
- **The pulmonologist concludes that the patient has lung-cancer currently at stage 0 and the top lining of the lung, documents it and sends it over to the nano team.**
- **Seventh, the patient is called to a medical room and asked to sit down on a bed for the pulmonologist to inject them with the nanobots.**
- **The pulmonologist injects the nanobots into the top lining of the lung which causes the patient minor to mild pain at first because of the needle.**
- **3-4 hours later, the medical device for transmission receives data from the nanobots who dissolve inside the patient.**
- **Eighth, the patient is then asked to come back to the next day where T-Cells are extracted from their body.**
- **The patient is then asked to come back for another appointment in a couple of days to give time for the nano team and the doctor to perform the training layer.**
- **Ninith, the patient returns and is injected with the trained white cells providing them with long-term immunity. The patient is discharged 2-3 days later as no side-effects were seen, and the immune system did not reject the trained T-cells.**

Cancer Treatment Trade-Offs

- **This cancer treatment framework has trade-offs just like any other system I have created and designed.**
- **I have identified 6 main trade-offs within the cancer treatment framework.**
- **The first trade-off is development cost.**
- **This trade-off exists because creating nanotechnology and combining it with immunotherapy can be extremely expensive and time-consuming.**
- **This trade-off can be managed through budget management, private funding and using pre-existing nano and immunotherapy systems.**
- **The second trade-off is false positives.**
- **This trade-off exists as sometimes scans or doctors can miss certain symptoms or signs of a different disease which isn't' cancer.**
- **This trade-off can be managed by the verification step, rotation schedules for medical staff and teams of people verifying results.**
- **The third trade-off is side-effects**
- **This trade-off exists as nanobots, or modified T-cells can cause unexpected responses from the immune system or other biological processes within the body.**
- **This trade-off can be managed through testing, monitoring the patient and adapting nano bots based on prior medical issues patients may have.**
- **The fourth trade-off is technical error.**
- **This trade-off exists as nanobots while inside the body can fail causing failed cancer treatments or no transmission of data.**
- **This trade-off can be managed by testing of nanobots, verification before deployment, biodegradable polymer material to prevent further issues, and groups of nanobots for redundancy.**
- **The fifth trade-off is human error.**
- **This trade-off exists as humans can be stressed, exhausted, scared or anxious which is common and perfectly healthy human behaviour.**
- **This means that doctors can make mistakes during detection or certain stages in the training or curing.**
- **This trade-off can be managed through rotation schedules, automated systems as a SafetyNet, fair & reasonable work hours and organised teams for each stage.**
- **The sixth and final trade-off is treatment cost.**
- **This trade-off exists as training, deploying nano bots, medical equipment, hiring medical staff, and automated systems can be expensive and time consuming to create and maintain.**
- **This trade-off can be managed by production teams, mass production of nanobots and budget management to manage costs.**
- **It is important to remember that there will be multiple implementation challenge especially with the nanobots, but it is not impossible.**